Clinic DBMS: Track B Integration Guide

DB leader ke liye: changes, naye phases aur fixes ko jaldi se system mein kaise jodna hai

1. Abhi system kahan tak pahuncha hai

Phase 0 se Phase 3 tak ka database kaam (P2 schema, P3-1 se P3-7 logic aur security, aur P3-8 tests) likha ja chuka hai. Poora schema aapke ER model (docs/design/01_er_model.md, rev 2) ke 103 entities se banta hai (+1 documented extension table). IMPORTANT: yeh sab abhi real MySQL par chala nahi hai, kyunki jahan bana wahan MySQL nahi tha. Static verifier pass hai, par pehli run mein 5-15 chhoti SQL errors expected hain.

Kya	Status	Kahan
Schema: 104 tables, 13 late FKs	Generated, run baaki	db/schema/*.sql (tools/schema_spec.py se)
P3-1 Foundation (352 triggers, approvals, alerts, login)	Likha	db/logic/01_foundation_triggers.sql
P3-2 Patients, scheduling, visits, stay, vitals	Likha	db/logic/02_patients_visits.sql
P3-3 Allergy, prescriptions (versions), orders, outside labs	Likha	db/logic/03_prescriptions_labs.sql
P3-4 Stock, dispensing, counter sale, custody, doses, leftovers	Likha	db/logic/04_pharmacy.sql
P3-5 Billing, payments, refunds, discharge, dues	Likha	db/logic/05_billing_discharge.sql
P3-6 Alerts, owner summary, exception reports, family links	Likha	db/logic/06_owner_alerts_links.sql
P3-7 Roles, accounts, grants (181 procedure grants, 58 views)	Generated	db/security/01_roles_and_grants.sql
P3-8 Rule tests: 127 tests + forbidden-action script	Likha, run baaki	db/tests/
Phase 4, 5, 6, 7	Aage	INTEGRATION_CONTRACT.md follow karo

2. Quick start (pehli baar)

```
python tools/verify_kit.py            # 'all checks passed'
cp .env.example .env                  # password set karo
docker compose down -v && docker compose up -d
bash tools/run_sql_checked.sh         # pehli error par ruk jaata hai
python tools/bootstrap_owner.py owner 'StrongPass' Asha Verma > owner.sql
docker compose exec -T mysql sh -c \
 'mysql -uroot -p"$MYSQL_ROOT_PASSWORD" clinic_db' < owner.sql
docker compose exec -T mysql sh -c \
 'mysql -uroot -p"$MYSQL_ROOT_PASSWORD" clinic_db' < db/tests/rule_tests.sql
bash db/tests/forbidden_actions.sh    # har line PASS honi chahiye
```

Rule tests ka output: har test ka PASS/FAIL aur rule number. FAIL aaye to us rule wali file fix hogi (section 3 ka impact map madad karega). Gate 2 tab, jab run_sql_checked.sh zero errors de. Gate 3 tab, jab saare 127 tests PASS hon.

3. Agar P1-3 / P1-4 (ya review) se design change aaye

Golden rule: schema ka single source of truth tools/schema_spec.py hai. Generated files (db/schema/*, db/logic/01_*) kabhi hand se edit nahi hoti.

Step by step

- python tools/compare_design.py <unka 02_table_design.md> -> dikhata hai kaun si tables sirf unke doc mein hain aur kaun si sirf hamare spec mein.
- Har farak par faisla: (a) unka sahi hai -> spec badlo. (b) hamara sahi hai -> CR likho. Dono ko alag mat chalne do.
- python tools/impact_map.py <table> -> us table ko chhune wale saare procedures, views, triggers aur tests ki list. Wahi aapka to-do hai.
- Spec badlo, phir: python tools/build_db.py, python tools/verify_kit.py, bash tools/run_sql_checked.sh, rule tests.
- Agar naya table/column kisi procedure ka hissa banna hai: procedure edit karo (db/logic/0X), @api comment update karo, python tools/gen_db_api_docs.py aur python tools/gen_security.py chalao.
- Naya rule test: tools/gen_rule_tests.py mein test() line jodo, regenerate, run.

Change ka type -> kya karna hai

Change	Kahan badlo	Kya re-run karo
Naya column (nullable)	schema_spec.py ke us table ki cols list	build_db, verify, run_sql_checked
Naya NOT NULL column	schema_spec + har INSERT jo us table mein hota hai (impact_map dikhayega)	build_db, verify, tests
Column rename	schema_spec + impact_map ki saari files (search/replace) + tests	verify_kit column checker pakdega
Naya table	schema_spec (table + cluster + rules) ; phir procedure/view	build_db, gen_security, gen_db_api_docs
Table ko tod ke normalize karna (P1-4 decomposition)	schema_spec mein naye tables; purani table ki jagah views banao jo purane column naam dete hain	impact_map se saare callers; tests
Enum/status naya value	schema_spec ka E(...) + related procedure ke checks	build_db, tests
CHECK / unique rule badla	schema_spec checks/uniques	build_db, tests
Naya business rule (R-number)	trigger snippet (bi/bu) ya procedure + test	verify, tests
Role/permission badli	procedure ke @api roles=... ; view access: tools/gen_security.py	gen_security, forbidden_actions.sh
Policy value (limit, days)	clinic_settings seed (schema_spec SEEDS) ya sp_set_setting	tests

P1-4 (normalization) se aksar yeh nikalta hai: koi table BCNF/4NF mein nahi hai. Aise me decomposition ko spec mein dalo, aur purani table ke naam par ek view rakho taaki baaki code na toote. Ek hi baar mein ek change karo aur har baar verify_kit chalao.

4. Agar Phase 4 / 5 / 6 / 7 ka kaam koi aur karke de

Jis AI se bhi yeh kaam karwao, uske prompt mein pehle docs/pooling/INTEGRATION_CONTRACT.md paste karo. Usme saare niyam hain: kaun se tables aur procedures use karne hain, connection kaise set karni hai, aur kya deliver karna hai.

Phase	Integration ke liye check	Common galti
P4-1 seed	Sirf procedures se data; staff create_staff se; price items CONSULT aur STAY_DAYCARE sp_set_price se	Seed mein seedha INSERT (triggers rok denge)
P4-2 bulk	Alag database clinic_bench, sirf schema files, triggers ke bina; recorded_by aur role khud bharo	Schema aur logic dono load karna (bahut slow)
P4-3 benchmarks	Index sirf benchmark ke liye alag script mein; schema_spec tab badlo jab index permanent ho	Production schema ko chupke se badalna
P4-4 showcase	Read-only SQL, tables/views baseline se	Naye column naam invent karna

Phase	Integration ke liye check	Common galti
P5 web app	app/routes/<naam>.py; DB calls call(conn, "sp_x", ...) helper se; roles clinic_<role>; @app_user_id/@app_role har request	Python mein rule dobara likhna
P6 demos	Procedure aur rule numbers docs/db_api aur rule_tests se	Purane naam use karna

Integration checklist (kaam wapas aane par)

- Files ko sahi folders mein rakho (har task sirf apni files likhta hai).
- python tools/verify_kit.py (web app ke CALL names, argument counts aur views bhi check hote hain).
- bash tools/run_sql_checked.sh phir rule tests: kuch toota to turant pata chalega.
- Agar AI ko kuch API mein nahi mila aur usne workaround kiya: use CR likhwao, workaround reject karo.
- Commit message mein task ID. Phir mujhe files + HANDOFF NOTE bhejo; main conflicts pool karunga.

5. Sabse kaam ke tools (ek line mein)

Command	Kaam
python tools/build_db.py	schema_spec se schema, late FKs, triggers aur baseline design doc banata hai
python tools/verify_kit.py	ER match, FK, columns (tables+views), CALL arguments, templates, Python syntax
python tools/impact_map.py [table]	Kis table ko kaun chhuta hai (change ka to-do)
python tools/compare_design.py <file>	Dusre design doc ko hamare spec se compare
python tools/gen_db_api_docs.py	docs/db_api/*.md ko SQL ke @api comments se banata hai
python tools/gen_security.py	roles, accounts, grants, security matrix, forbidden-action script
python tools/gen_rule_tests.py	db/tests/rule_tests.sql (127 tests)
bash tools/run_sql_checked.sh	Sab SQL files load; pehli error par ruk ke file batata hai
python tools/bootstrap_owner.py	Pehla owner account
python tools/make_guide.py	Yeh guide (md + pdf) dobara banata hai

6. Pehli run mein expected errors

Error	Matlab / fix
1418 DETERMINISTIC / NO SQL / READS SQL DATA	compose flag --log-bin-trust-function-creators=1 nahi laga: docker compose down -v phir up
3780 / 1215 FK type mismatch	schema_spec mein dono columns ka type same karo
1064 syntax error near ...	Line number aur file mujhe do. Generated ho to tools/ mein fix
1442 cannot update table in trigger	Trigger usi table ko modify kar raha hai; mujhe batao
1305 PROCEDURE/FUNCTION does not exist	Load order: run_all.sql ki order ya naam typo; verify_kit ye pehle pakadta hai
Test FAIL: no error was raised	Wo rule enforce nahi ho raha; rule number se file dhundo
Test FAIL: Duplicate / Not enough stock	Pehle ke run ka data; db/reset.sql + run_all dobara

7. Safety

- Real patient data, .env ya password kisi AI mein paste mat karo. DEV passwords (Dev#Owner2026 jaise) sirf local demo ke liye hain: ALTER USER se badlo.
- owner.sql commit mat karo. Koi AI bole "tests pass", aap khud chalao.

- Gate tags: design-v1 (aapka Phase 1), schema-v1 (run_sql_checked zero errors), logic-v1 (127 tests PASS).

Appendix: design decisions (details: docs/pooling/CR_LOG_TRACK_B.md)

- CR-2: merge event rows ko rewrite nahi karta; patient_merge_log redirect hai (canonical_patient()).
- CR-B1: staff_credential extension (password hash). CR-B2: message_outbox.rendered_text extension.
- CR-B4: password hashing scrypt (stdlib). CR-B10: ek visit ka ek bill; counter sale ka alag payment/refund path.
- CR-B12: galat patient ki report store nahi hoti; message wapas aata hai aur interrupt alert bachta hai.
- CR-B14: emergency override sirf paise wala step chhodta hai; sign-off, doses, reports phir bhi chahiye.